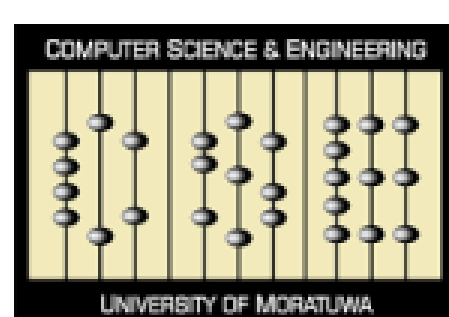




Patient-Aware Time-to-Event Modeling of Atrial Fibrillation Onset using Heart Rate Variability

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Abstract

Atrial fibrillation (AF) is a prevalent cardiac arrhythmia carrying significant stroke risk, making early personalised prediction clinically valuable. We propose a patient-aware contrastive learning framework that learns discriminative sinus rhythm (SR) and AF representations from RR interval sequences, restricting positive pairs to same-patient segments to preserve individualised rhythm dynamics. A patient-specific SR–AF axis in the latent space then yields an interpretable, continuous measure of AF progression suitable for edge deployment.

Project Objectives

- Learn discriminative SR vs. AF representations via patient-aware contrastive learning
- Estimate time-to-AF onset without labelled deviation points during training
- Validate progression scoring on unseen patients via SR–AF axis projection
- Enable edge-deployable inference from RR interval sequences

Related work

Classical AF detection stacks hand-crafted HRV features with shallow classifiers [1]. Deep models on RR intervals [2] mainly pursue AF vs. SR classification and give limited emphasis to onset timing and patient heterogeneity. HRV trends anticipate onset [3]; survival models address time-to-event; standard contrastive SSL [4] weakly preserves patient-specific cardiac dynamics without constrained positives.

Methodology

Encoder Architecture

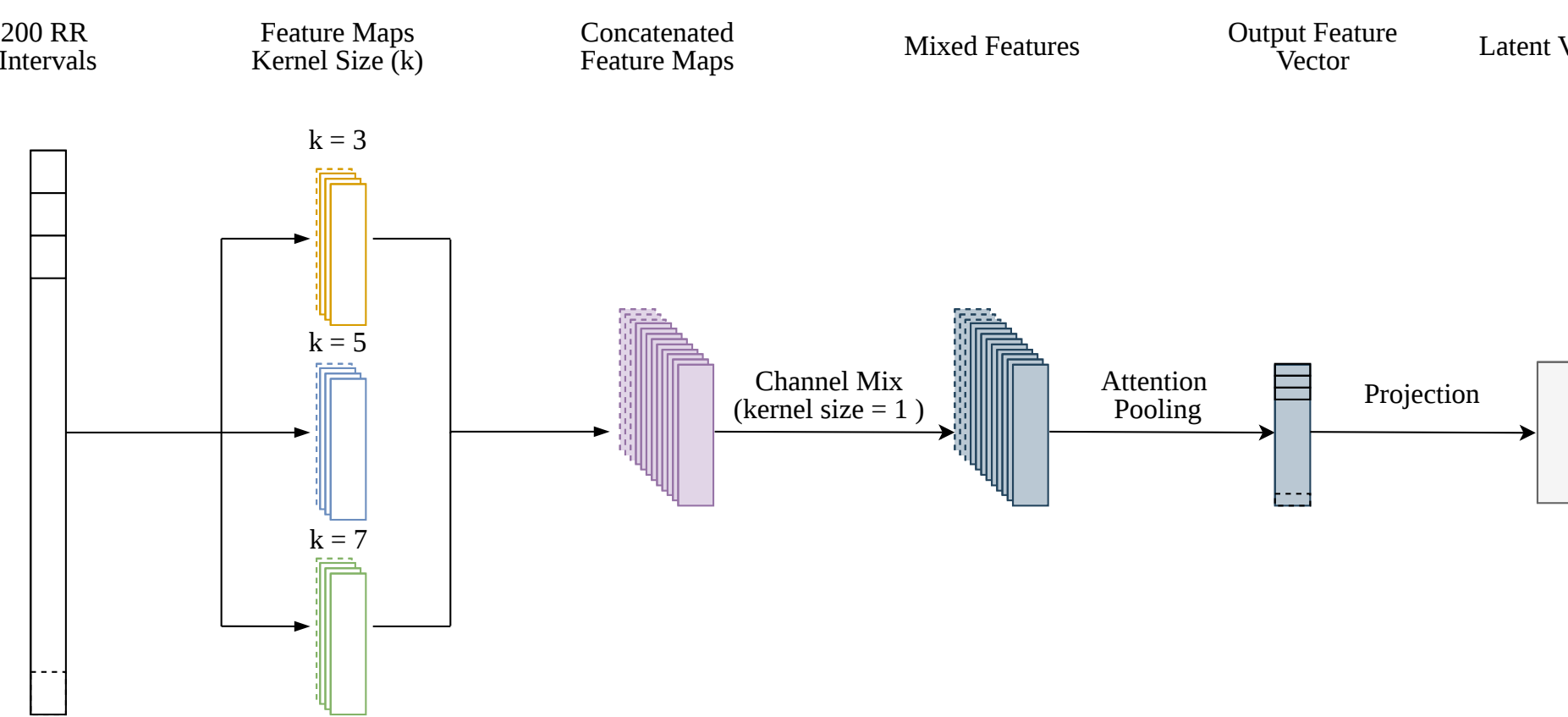


Figure 1. Multi-branch convolutional backbone with temporal attention pooling and a two-layer MLP projection head for patient-aware contrastive learning.

- Per-patient robust norm on SR RR segments
- ℓ_2 -normalised 2-layer MLP projection head
- Patient-aware contrastive — same patient, same class positives

SR–AF Axis Projection

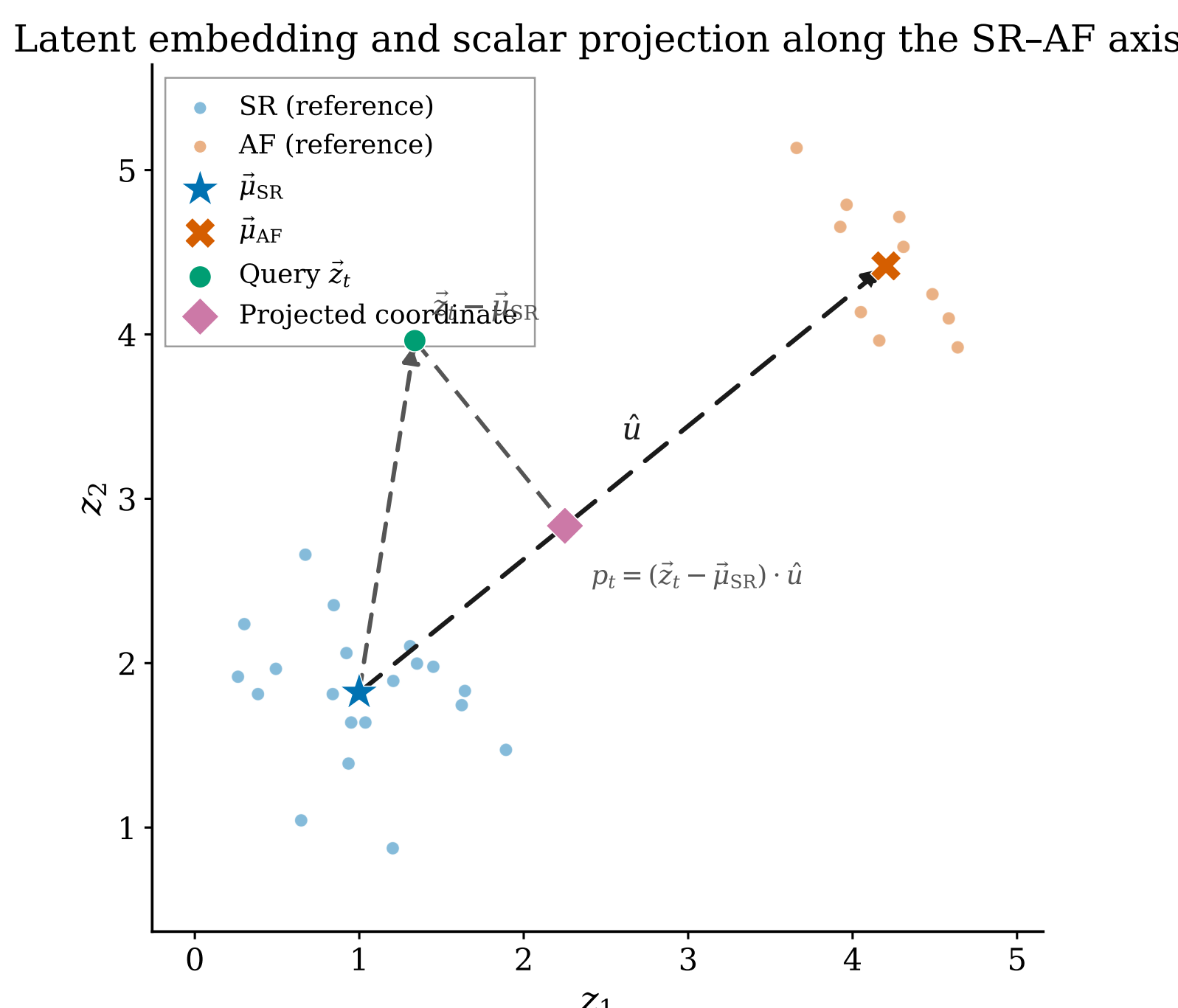


Figure 2. SR–AF axis per patient in frozen latent space; $p_t \in [0, 1]$ between μ_{SR} and μ_{AF} along axis.

Project latent embeddings onto patient SR→AF axis:

$$p_t = \frac{(z_t - \mu_{SR}^{(p)}) \cdot (\mu_{AF}^{(p)} - \mu_{SR}^{(p)})}{\|\mu_{AF}^{(p)} - \mu_{SR}^{(p)}\|^2} \quad (1)$$

- p_t : 0↔SR centroid; 1↔AF centroid along axis

Experiments

Dataset

Source IRIDIA-AF — 167 patients (paroxysmal AF); RR intervals from long-term single-lead ECG

Episode criteria AF episode ≥ 1 h; preceding SR ≥ 4 h; episode fully inside one continuous recording

Split Strict patient-level train / val / test: 119 / 24 / 24 patients $\Rightarrow$ 154 / 27 / 24 episodes

Evaluation Protocol

- Linear probe:** Frozen encoder; logistic regression on train embeddings $\rightarrow$ held-out patients (test)
- Baselines:** Identical encoder and protocol; SupCon vs. BCE

Results

Per-Class Performance (Proposed Loss)

Class	Precision	Recall	F1
SR	0.970 ± 0.009	0.927 ± 0.020	0.948 ± 0.012
AF	0.936 ± 0.017	0.974 ± 0.008	0.955 ± 0.010
Accuracy	0.952 ± 0.011		
AUROC	0.989 ± 0.003		

Table 1. Mean $\pm$ std across 5 seeds; 4160 AF and 3799 SR segments from unseen patients.

Headline metrics (ours)

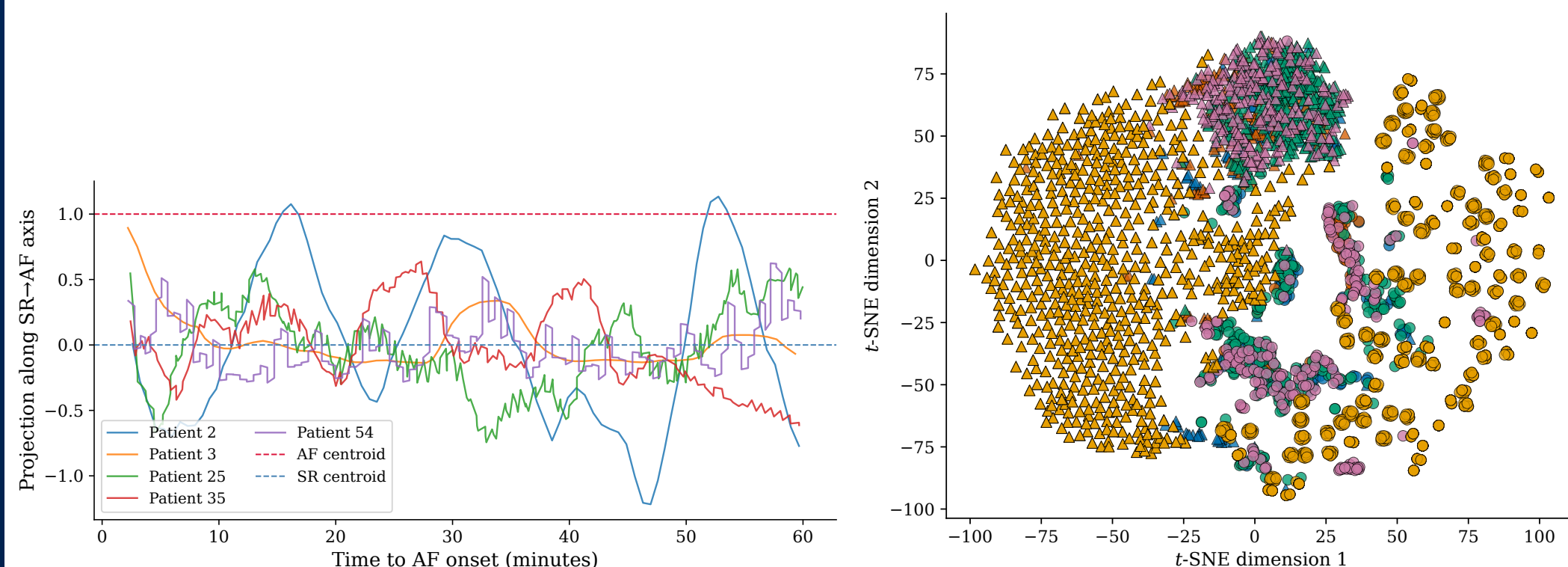
AUROC **0.989 ± 0.003** AF recall **0.974 ± 0.008** Accuracy **0.952 ± 0.011**

Latent Space Quality (Test Split)

Method	SR Spread $\uparrow$	AF Cohesion $\uparrow$	Centroid Sim.
BCE	0.478	0.962	−0.740
SupCon	0.348	0.968	0.176
Ours	0.551	0.850	0.266

Table 2. SupCon's tight SR cohesion indicates inter-patient SR collapse; Ours preserves individual SR manifolds.

Qualitative Analysis — Five Held-Out Test Patients



(a) Patient-specific SR–AF axis projections (p_t). (b) t-SNE of encoder embeddings coloured by rhythm class.

- p_t : interpretable progression score toward AF along patient axis
- t-SNE: patient-specific SR clusters preserved

Conclusions

- Patient-aware contrastive learning reliably detects paroxysmal AF, achieving **AUROC 0.989 ± 0.003** on strictly unseen patients
- Same-patient positives prevent SR prototype collapse; inter-patient rhythm variability retained
- Outperforms SupCon and BCE on AUROC and stability (**up to $3.4\times$ lower seed variance**)

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